

Digital Forensics & Incident Response

UNIT 2 - Digital Evidence Acquisition & Data Recovery

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Digital Evidence Acquisition and Preservation



Digital Evidence Acquisition and Preservation

Digital evidence acquisition is the process of obtaining relevant digital information from a device or system for forensic examination while maintaining its integrity.

The objective is not simply to copy files. The investigator must acquire evidence in a manner that allows the resulting copy to be examined without unnecessarily modifying the original source.

Acquisition May Include

- Physical storage devices
- USB drives
- Forensic disk images
- Volatile memory
- Relevant system information

Main Objective

Acquire → Preserve → Verify → Analyse



Evidence Preservation

Evidence preservation means protecting digital evidence from alteration, destruction, contamination or unauthorised access during an investigation.

Digital evidence is particularly sensitive because normal computer operations can modify stored information. For example, simply starting a computer may change certain timestamps or write information to the storage device.

Preservation therefore requires:

- Controlled evidence handling
- Appropriate acquisition procedures
- Integrity verification
- Proper documentation
- Protection of the original evidence



Write Blockers

A **write blocker** is a hardware or software mechanism designed to prevent data from being written to a storage device while allowing the investigator to read and acquire information from it.

The primary purpose is to protect the original evidence from accidental modification during forensic examination.

Basic Concept

Suspect Drive



Write Blocker



Forensic Workstation



The investigator can access the contents while reducing the possibility of changes being written back to the original device.



What Can Happen Without a Write Blocker?

If a suspect storage device is connected directly to a computer and the operating system interacts with it, information may potentially be changed.

Activity such as:

- Access timestamps
- Swap-file activity
- Last-mounted information

For example, booting from or accessing a suspect drive may cause the operating system to perform operations that alter information on the drive.

Forensic Principle

The original evidence should be protected from unnecessary modification.

This is why write-blocking mechanisms are important during forensic acquisition.



Hardware vs Software Write Blockers

Hardware Write Blocker

A hardware write blocker is a **physical device** placed between the evidence storage device and forensic workstation. It is designed to prevent write commands from reaching the original device.

Software Write Blocker

A software write blocker uses software controls to prevent or restrict writing to the evidence media.

Hardware	Software
Physical device	Software-based control
Positioned between device and workstation	Implemented through the forensic environment
Provides dedicated write protection	Depends on correct software configuration
Commonly used for forensic acquisition	Useful where appropriate software controls are available



Forensic Imaging



Forensic Imaging

Forensic imaging is the process of creating a forensic copy of a storage device or other digital evidence source for examination. Instead of performing the investigation directly on the original storage device, investigators can work with the forensic image.

Why Create an Image?

A forensic image can preserve a representation of the source that can be:

- Examined repeatedly
- Verified using hashes
- Analysed with forensic tools
- Stored as evidence
- Shared with authorised investigators

Important Principle

Original evidence → Protected

Forensic image → Examination



FTK Imager

What is FTK Imager?

FTK Imager is a forensic imaging and evidence-preview tool used to acquire and examine digital evidence. It is used for creating forensic images, recording case information, calculating hashes and verifying the resulting image.

Example: creating a **physical drive image in .E01 format** using FTK Imager.

Major Uses in This Unit

- Creating forensic images
- Recording case information
- Calculating hashes
- Verifying images
- Viewing evidence without modifying the source



Forensic Image: E01 Format

E01 Forensic Image

The **E01 format** is a commonly used forensic image format associated with Expert Witness-style evidence files. Instead of simply copying selected files, forensic imaging can preserve a much broader representation of the source storage media.

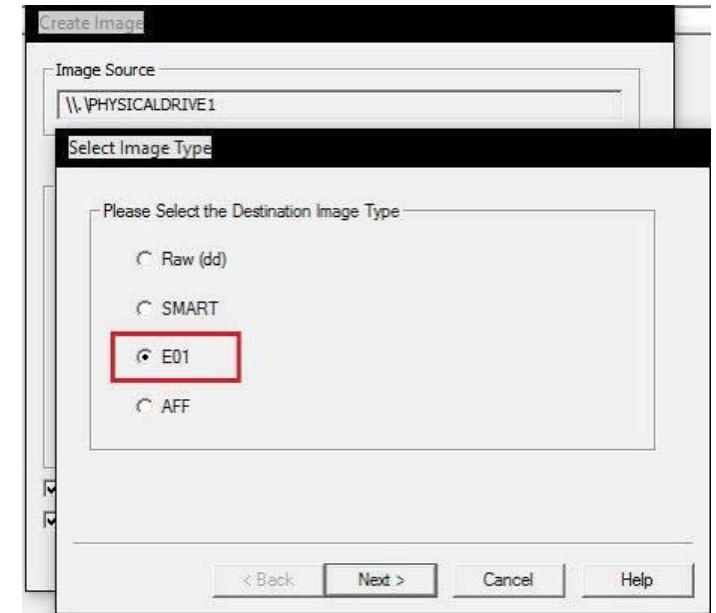
During Imaging

Investigators can associate the image with:

- Case information
- Evidence information
- Examiner details
- Acquisition information
- Integrity information

Purpose

The resulting image becomes a controlled source for subsequent forensic examination.



Hashing in Digital Forensics

A **cryptographic hash** is a fixed-length value generated from digital data using a mathematical algorithm.

A change in the underlying data should result in a different hash value.

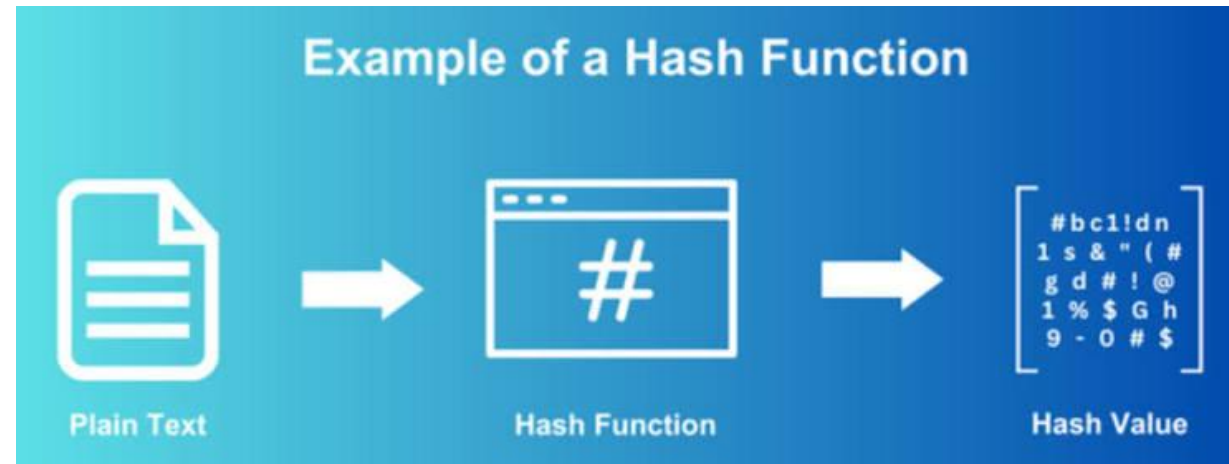
Common Algorithms in This Unit

MD5

SHA-256

Hashes can be calculated for:

- Original evidence
- Forensic images
- Files
- Acquired data



The values can then be compared to determine whether the data being examined matches the previously calculated value.



Evidence Integrity Through Hashing

Consider a forensic image with the following hash:

SHA-256: A1B2C3...

Later, the investigator calculates the SHA-256 hash again.

If the values match: The data is consistent with the previously hashed version.

If the values differ: The data has changed in some way, and the investigator should investigate the reason.

Simple Flow

Evidence

- Hash Calculation
- Forensic Image
- Hash Verification
- Compare Values

Key Principle

Hashing provides a mechanism for verifying data integrity.



Image Verification

Image verification is the process of checking that a forensic image accurately corresponds to the acquired evidence by comparing integrity information such as hash values.

After imaging, the forensic tool can generate a verification result.

Verification Process

Source Evidence



Create Forensic Image



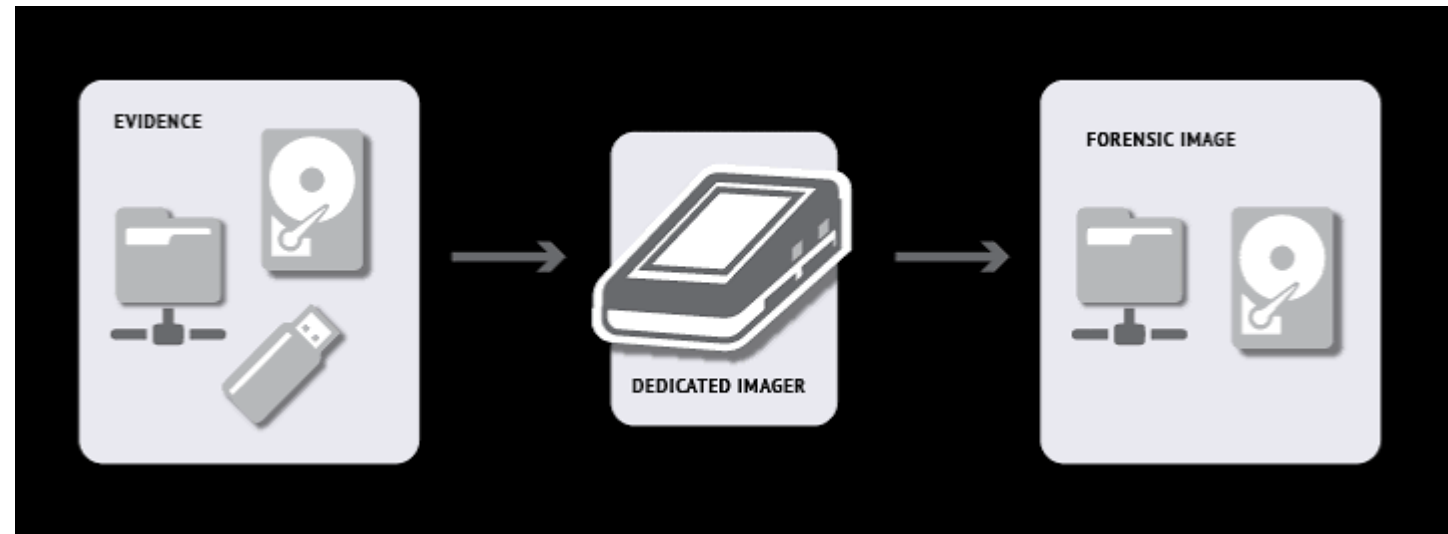
Calculate / Record Hash



Verify Image



Compare Hash Values



Result

A successful verification provides evidence that the image remained consistent with the acquisition process.



Viewing Evidence Without Modifying the Source

Forensic investigators should avoid conducting analysis directly on original evidence whenever possible. Instead, the forensic image can be mounted or loaded into an investigation tool in a manner that allows examination without writing changes back to the original source.

This allows investigators to:

- Browse files
- Examine file systems
- Search for keywords
- Identify deleted files
- Build timelines
- Generate reports

Core Principle

Acquire first → Analyse the forensic copy

rather than repeatedly interacting with the original evidence.



Introduction to Autopsy



Introduction to Autopsy

What is Autopsy?

Autopsy is a digital forensics platform used to examine forensic evidence and investigate digital artefacts. It is introduced for analysing forensic images and recovering relevant information.

Major Activities

- Creating a forensic case
- Adding an .E01 data source
- Running ingest modules
- Navigating the file tree
- Recovering deleted files
- Searching for keywords
- Viewing timelines
- Generating reports



Autopsy Case Setup

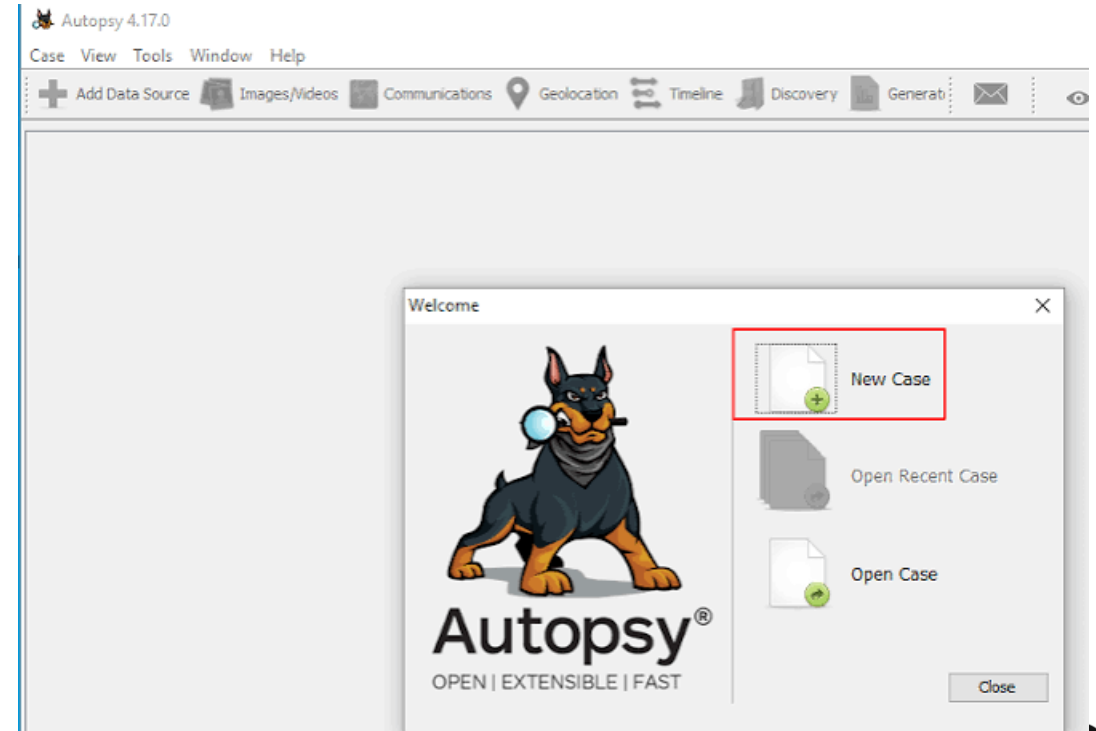
A forensic investigation in Autopsy begins by creating a **case**.
The case provides a structured environment for organising evidence and investigation results.

Typical Process

**Create Case → Enter Case Information → Add Data Source →
Run Ingest Modules → Examine Artefacts → Analyse Findings
→ Generate Report**

Importance

Organising evidence within a case helps maintain a structured investigation rather than examining individual files without context.



Ingest Modules in Autopsy

What are Ingest Modules?

Ingest modules are components within Autopsy that process forensic data and identify useful information for investigation. They can assist with analysing different categories of evidence.

Purpose

Instead of manually examining every piece of data, ingest processing can identify and organise potentially relevant information.

Investigation Examples

- File-system information
- Deleted files
- Keyword matches
- Other forensic artefacts
- Timeline-related information



Deleted File Recovery

What is Deleted File Recovery?

Deleting a file does not necessarily mean that all information associated with that file has immediately disappeared from the storage device.

File-system information may indicate that a file is deleted while some or all of its underlying data may remain available for recovery.

In Autopsy

The syllabus specifically introduces recovery of files marked as **unallocated in the MFT (Master File Table)**.

Important Concept

Deleted $\neq$ Immediately Destroyed

Recovery depends on factors such as whether the storage space has been reused or overwritten.



Keyword Search in Forensics

Keyword searching is the process of searching forensic evidence for specific words, phrases or patterns that may be relevant to an investigation.

For example, an investigation involving suspected data theft may search for:

- Project names
- Confidential document names
- Usernames
- Email addresses
- Specific terms

Why is it Useful?

Large forensic images may contain thousands or millions of files. Keyword searching helps investigators **narrow the investigation to potentially relevant information**.



File Carving, Magic Bytes & Slack Space Mechanics

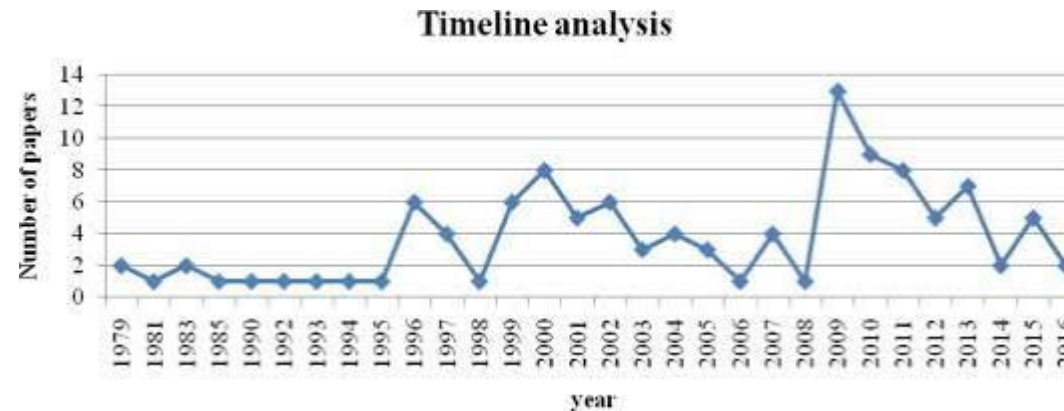


Timeline Analysis

A **forensic timeline** organises relevant events according to their associated timestamps. Timeline analysis helps investigators understand **the sequence in which activities occurred**.

Example

09:10 — User logged in
09:15 — File accessed
09:18 — USB device connected
09:20 — File modified
09:25 — USB device removed



Why is it Useful?

A timeline allows investigators to correlate individual artefacts and develop a chronological understanding of an incident.



File Carving

File carving is a data-recovery technique used to identify and reconstruct files from raw storage data without relying completely on the file system's directory information.

It is particularly useful when file-system metadata is missing or when attempting to recover deleted files.

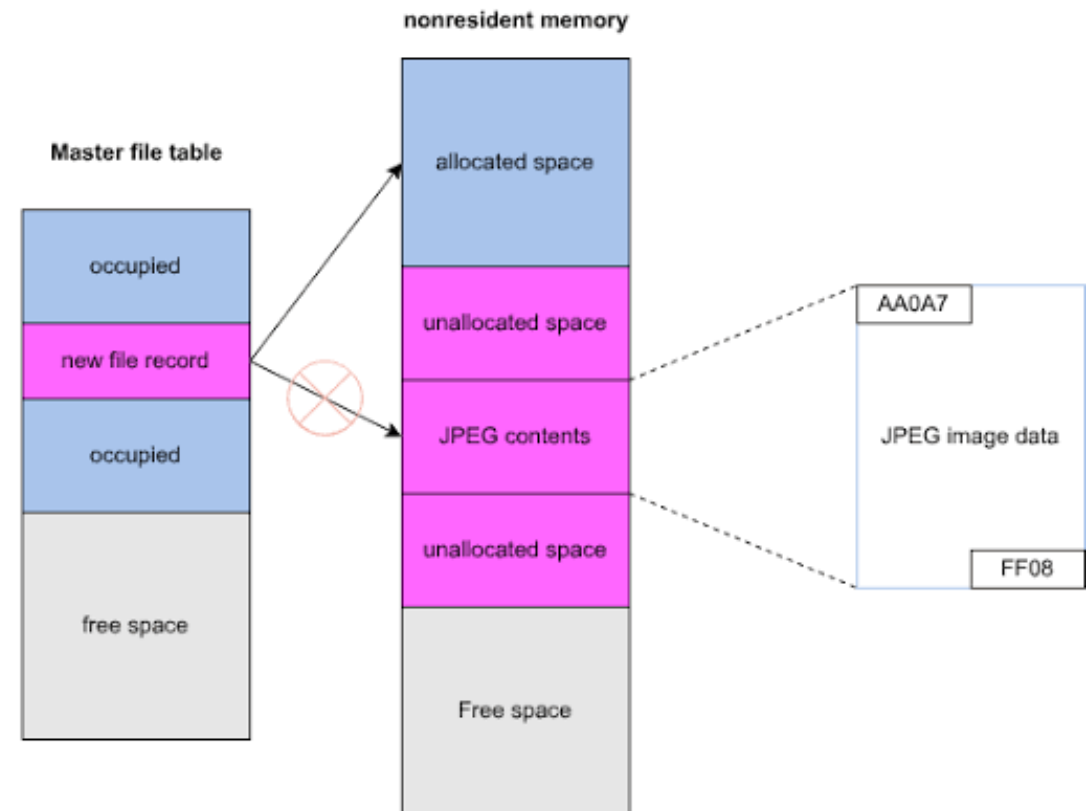
Basic Principle

File carving can use **file signatures** to identify the beginning or structure of known file types.

Example

A JPEG file has characteristic identifying bytes.

An investigator can search storage data for those patterns and attempt to reconstruct the associated file.



File Signatures and Magic Bytes

A **file signature**, often called **magic bytes**, is a characteristic sequence of bytes associated with a particular file format. These bytes can help forensic tools identify the type of data contained in a section of storage.

Concept

Instead of relying only on:

document.jpg

the investigator can examine the underlying binary data to identify whether it actually corresponds to a JPEG file.

Why Useful?

- File signatures are particularly important when:
- File names are missing
- File extensions are incorrect
- File-system information has been damaged
- Deleted files need to be recovered



Slack Space

Slack space is the unused space within the final allocated cluster of a file.

A file may not completely fill the storage cluster allocated to it. The remaining portion is called slack space.

Example

Suppose a storage cluster has:

4 KB capacity

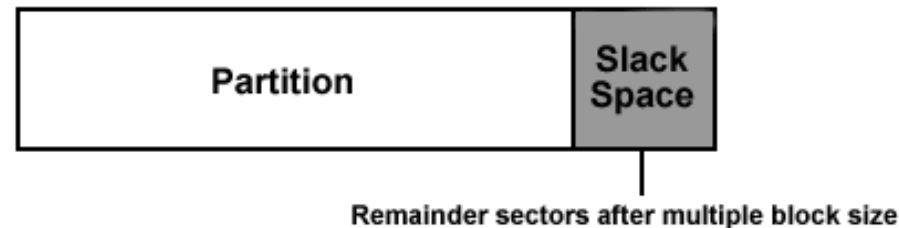
but a file uses:

3 KB

The remaining:

1 KB

may exist as slack space.



Forensic Importance

Slack space can sometimes contain remnants of previously stored information, depending on the storage and file-system conditions.

The syllabus specifically defines slack space as the space between the end of file data and the end of its cluster.



Unallocated Space

Unallocated space is storage space that is not currently assigned to an active file according to the file system. When a file is deleted, the file system may mark the associated space as available for future use rather than immediately erasing every underlying byte.

Why is it Important?

Unallocated space may contain remnants of:

- Deleted files
- File fragments
- Previous data
- Other recoverable information

Forensic Principle

Unallocated does not necessarily mean empty.

This is one of the important concepts behind deleted-file recovery and file carving.



Swap Space

Swap space is storage space used by an operating system as an extension of memory when required. Depending on the operating system, it may contain information that was previously held in memory.

Forensic Relevance

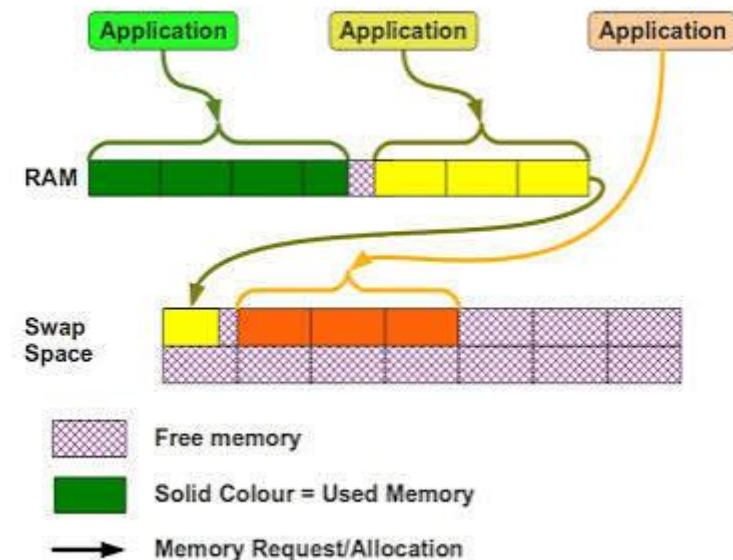
Swap data can potentially contain fragments of information associated with system activity. Therefore, forensic examination may consider swap space as another potential source of evidence.

Important Distinction

RAM: Primarily volatile memory

Swap: Storage-based space that may contain memory-related information

Disk: Persistent storage containing files and other artefacts



Memory Forensics & Volatility 3 Plugin Execution



Volatility 3

What is Volatility 3?

Volatility 3 is a memory-forensics framework used to analyse captured memory dumps.

The Unit II syllabus introduces Volatility 3 for identifying the operating system, examining running processes and analysing network connections.

Key Plugins in This Unit

windows.info

Used for Windows memory-image and operating-system identification.

windows.pslist

Used to examine running processes.

windows.netstat

Used to examine network connections.



Windows.info Plugin

Purpose: The **windows.info** plugin helps identify information about the Windows environment represented by the memory dump.

Before performing detailed memory analysis, an investigator needs to understand the operating-system context of the captured memory.

Why Important?

Memory structures and analysis techniques depend on the operating-system environment.

Investigation Flow

Memory Dump



windows.info



Identify OS Information



Select Appropriate Analysis



Windows.pslist Plugin

Purpose: The **windows.pslist** plugin is used to examine processes present in a Windows memory image.

Why Analyse Processes?

Running processes provide information about what was executing on the system when the memory was captured.

Investigators may look for:

- Unusual process names
- Suspicious parent-child relationships
- Unexpected processes
- Processes associated with malicious activity

Important Point

A suspicious process is an **indicator requiring further investigation**, not automatically proof of malicious activity.



Windows.netstat Plugin

Purpose: The **windows.netstat** plugin can be used to examine network connections represented in a Windows memory image.

Investigators may examine:

- Local addresses
- Remote addresses
- Ports
- Connection state
- Associated processes, where available

Forensic Value

Network information from memory can help investigators identify potentially suspicious communication and correlate it with running processes or other evidence.



Memory Forensics Investigation Flow

Memory Analysis with Volatility 3

Memory Dump



Identify Operating System

windows.info



Analyse Running Processes

windows.pslist



Analyse Network Connections

windows.netstat



Correlate Findings



Investigate Suspicious Activity

The purpose is to reconstruct aspects of the system's state **at the time the memory was captured.**



Integrated Example: Recovering Digital Evidence

Scenario: A computer is suspected of running malware and is still powered on.
The investigator considers the possibility that important information may disappear if the system is shut down.

Investigation Considerations

Volatile Evidence

RAM and active network connections may be time-sensitive.

Memory Acquisition

A memory capture may preserve information about the running system.

Volatility 3

The memory dump can be analysed using:

windows.info → OS identification

windows.pslist → Running processes

windows.netstat → Network connections

Objective

Correlate memory evidence with other forensic evidence to understand the system's state and identify suspicious activity.



Thank you

